

RED TEAM DLP TEST REPORT

S3 Data Exfiltration Test Tool — Walkthrough & Analysis

Document Type	Technical Walkthrough Report
Classification	CONFIDENTIAL — Authorized Personnel Only
Test Tool	S3_DLP_RedTeam_Tool.exe
Test Type	Data Loss Prevention (DLP) Validation
Platform	Windows 10/11 (GUI executable)
Test Environment	Isolated / Authorized Test Scope Only
Prepared By	Red Team
Date	May 2026

⚠ This document and the tools described herein are for AUTHORIZED security testing only. Unauthorized use is illegal and a violation of organizational policy.

1. Executive Summary

This report documents the walkthrough and usage of S3_DLP_RedTeam_Tool.exe — a purpose-built Windows GUI application developed to perform authorized data exfiltration simulation tests against AWS S3 storage. The tool is designed to validate whether an organization's Data Loss Prevention (DLP) controls can detect and block unauthorized data transfers to cloud storage.

The tool simulates 10 distinct exfiltration techniques, ranging from simple file uploads to advanced bypass methods such as presigned URLs, encoded payloads, and renamed file extensions. All tests use synthetic, clearly labeled test data and target an authorized test S3 bucket.

Objective: Identify gaps in DLP coverage across CLI, GUI, and programmatic AWS S3 upload paths so that security controls can be tuned and hardened accordingly.

2. Tool Overview

2.1 What is S3_DLP_RedTeam_Tool.exe?

S3_DLP_RedTeam_Tool.exe is a standalone Windows executable (no installation required) that provides a graphical interface for running structured data exfiltration tests against a target AWS S3 bucket. It is built using Python (tkinter + boto3) and compiled with PyInstaller into a single portable file.

Property	Details
File Name	S3_DLP_RedTeam_Tool.exe
Platform	Windows 10 / 11 (64-bit)
Language	Python 3.12 (compiled to native exe via PyInstaller)
Dependencies	Bundled — no Python, no AWS CLI, no additional install needed
AWS SDK	boto3 (bundled)
Test Methods	10 distinct exfiltration scenarios
Output	On-screen results table + exportable CSV log

2.2 GUI Layout

The application window is divided into two panels:

Panel / Section	Purpose
AWS Credentials	Enter Access Key ID, Secret Access Key, and Region to authenticate with AWS.
Target Bucket	Specify the S3 bucket name and key prefix (folder path) for exfiltrated files.
Test Data	Generate synthetic PII/confidential files or browse for a custom file to upload.
Test Scenarios	10 checkboxes — select which tests to run. Use Select All / Deselect All buttons.

Controls	Run, Stop, Cleanup, and Export buttons to manage the test session.
Results Table	Live results: test name, file, upload status, DLP alert status, timestamp.
Activity Log	Timestamped console-style output showing every action and result in real time.
Progress Bar	Shows how many tests are complete out of total selected.

3. Step-by-Step Walkthrough

Step 1 — Launch the Tool

Double-click S3_DLP_RedTeam_Tool.exe. On first launch an Authorization Disclaimer dialog appears:

The disclaimer requires the tester to confirm written Rules of Engagement, isolated test environment, synthetic data only, and DLP team on standby — before proceeding.

Click "I Confirm — Proceed" to continue into the main application window.

Step 2 — Enter AWS Credentials

In the AWS Credentials panel (top-left):

- Enter the Access Key ID of the red team IAM test account
- Enter the Secret Access Key
- Set the Region (default: us-east-1)
- Click the ↗ Connect to AWS button

The connection status indicator changes to green (● Connected) when authentication succeeds. Any credential error is displayed immediately in the Activity Log.

Only the minimum required IAM permissions are needed: s3:PutObject, s3:DeleteObject, s3:ListBucket on the test bucket only.

Step 3 — Configure Target Bucket

In the Target Bucket panel:

- Enter the Bucket Name of the authorized test S3 bucket
- Set the Key Prefix (default: exfil/) — this is the S3 folder where test files land

Step 4 — Generate Synthetic Test Data

Click ☐ Generate Synthetic Test Data. The tool creates three files in a local temporary folder (~\redteam_dlp_test/):

File	Size	Contents
pii_data.csv	< 1 KB	Fake names, SSNs, credit card numbers, emails — triggers PII DLP rules
confidential.txt	< 1 KB	Marked CONFIDENTIAL with fake budget and strategy data — triggers keyword rules
bulk_50mb.bin	50 MB	Random binary data — triggers volume-based DLP thresholds

All files are clearly labeled RED TEAM DLP TEST ONLY — they contain no real sensitive information.

Step 5 — Select Test Scenarios

Check or uncheck the 10 scenario checkboxes based on what you want to test. All are selected by default. Use Select All / Deselect All for quick toggling.

Step 6 — Run the Tests

Click ► Run Selected Tests. The tool executes each selected test sequentially in a background thread, so the GUI remains responsive throughout:

- The Activity Log fills with timestamped entries showing each upload attempt
- The Results Table updates after each test with upload status
- The progress bar shows completion percentage

After all tests finish, the log shows SESSION COMPLETE. The DLP team should now review their console for triggered alerts and fill in the DLP Alert / Blocked columns in the results table.

Step 7 — Export Results

Click ☐ Export Results CSV to save the results table to a .csv file for inclusion in the formal DLP test report.

Step 8 — Cleanup

Click ☐ Cleanup All Artifacts to:

- Delete all uploaded S3 objects under the test prefix
- Remove the local ~/redteam_dlp_test/ folder

⚠ Always run cleanup before ending the session to prevent accidental data exposure and avoid ongoing AWS storage costs.

4. Test Scenarios — Detailed Description

The table below describes each of the 10 built-in test scenarios, the technique used, and the DLP detection it is designed to validate.

ID	Name	Method	Description & DLP Rule Targeted	Upload	DLP Expected
T-01	Single PII Upload	boto3	Uploads pii_data.csv directly. Targets PII content-inspection rules (SSN, CCN, email patterns).	✓ Success	Content alert
T-02	Bulk Sync	boto3	Syncs entire test folder to S3. Targets multi-file and volume-based DLP policies.	✓ Success	Volume alert
T-03	Large File 50 MB	boto3	Uploads a 50 MB binary file. Targets file-size threshold rules.	✓ Success	Size alert
T-04	Compressed TAR.GZ	boto3	Archives PII + confidential files into tar.gz before upload. Tests if DLP inspects inside compressed containers.	✓ Success	Archive bypass?
T-05	Multipart Upload	boto3	Splits 50 MB into 5 MB chunks via S3 multipart API. Tests if DLP reassembles and inspects chunked transfers.	✓ Success	Chunk bypass?
T-06	Presigned URL	urllib	Generates a presigned PUT URL, then uploads via HTTP with no AWS credentials on sender side. Tests unsigned-request DLP.	✓ Success	No-cred bypass?
T-07	Base64 Encoded	boto3	Encodes PII file as base64 before upload. Tests if DLP decodes and inspects encoded payloads.	✓ Success	Encode bypass?
T-08	Renamed Extension	boto3	Renames pii_data.csv to vacation_photo.jpg. Tests if DLP uses content inspection vs extension-only matching.	✓ Success	Ext bypass?
T-09	Password ZIP	boto3	Wraps PII in a ZIP file. Tests if DLP can inspect or block password-protected containers.	✓ Success	ZIP bypass?
T-10	Confidential TXT	boto3	Uploads confidential.txt with CONFIDENTIAL keyword and sensitive strategy data. Targets keyword/regex DLP rules.	✓ Success	Keyword alert

Legend: Blue = DLP expected to detect | Orange = Potential bypass — verify manually

5. DLP Validation Checklist

After running each test, the DLP team should complete the following checklist:

Test	Scenario	File Uploaded	DLP Alert Fired	Alert Time	Traffic Blocked	Notes
T-01	Single PII CSV	☐ Yes ☐ No	☐ Yes ☐ No		☐ Yes ☐ No	
T-02	Bulk Sync	☐ Yes ☐ No	☐ Yes ☐ No		☐ Yes ☐ No	
T-03	Large File 50 MB	☐ Yes ☐ No	☐ Yes ☐ No		☐ Yes ☐ No	
T-04	TAR.GZ Archive	☐ Yes ☐ No	☐ Yes ☐ No		☐ Yes ☐ No	
T-05	Multipart Upload	☐ Yes ☐ No	☐ Yes ☐ No		☐ Yes ☐ No	
T-06	Presigned URL	☐ Yes ☐ No	☐ Yes ☐ No		☐ Yes ☐ No	
T-07	Base64 Encoded	☐ Yes ☐ No	☐ Yes ☐ No		☐ Yes ☐ No	
T-08	Renamed Extension	☐ Yes ☐ No	☐ Yes ☐ No		☐ Yes ☐ No	
T-09	Password ZIP	☐ Yes ☐ No	☐ Yes ☐ No		☐ Yes ☐ No	
T-10	Confidential TXT	☐ Yes ☐ No	☐ Yes ☐ No		☐ Yes ☐ No	

6. Common DLP Gaps & Recommendations

Gap Found	Affected Tests	Recommended Fix
Presigned URL uploads bypass DLP	T-06	Enable CASB with API-level S3 visibility; inspect unsigned PUT requests
Compressed archives not inspected	T-04, T-09	Configure DLP to decompress and inspect TAR.GZ, ZIP containers
Base64 encoding bypasses content scan	T-07	Enable content normalization / decode-before-inspect in DLP engine
Renamed extension not caught	T-08	Switch from extension-based to magic-byte/content-based file identification
Chunked multipart not reassembled	T-05	Enable stream reassembly in proxy/DLP for multipart S3 transfers

Volume threshold too high	T-02, T-03	Lower volume alert threshold; alert on any S3 upload outside approved buckets
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7. Post-Test Cleanup Procedure

After every test session the following cleanup steps must be performed:

1. Click ☐ Cleanup All Artifacts in the tool — deletes all S3 objects under the test prefix and removes local test files
2. Verify S3 bucket is empty: AWS Console → S3 → check exfil/ prefix has no objects
3. Confirm local folder `~/redteam_dlp_test/` has been removed
4. Revoke or rotate the test IAM credentials if they were created specifically for this exercise
5. Provide the exported CSV results to the DLP team for alert correlation
6. Document any DLP gaps in the findings register

Never leave test data in S3 after a session. Even clearly labeled synthetic data should be removed to maintain a clean test environment.

8. Tool Architecture Notes

8.1 How the EXE is Built

The tool is written in Python and compiled to a standalone Windows executable using PyInstaller (`--onefile`). All dependencies including boto3, botocore, and tkinter are bundled inside the single .exe file.

Build command (run on Windows):

```
python -m PyInstaller --onefile --windowed --name S3_DLP_RedTeam_Tool \
  --collect-all boto3 --collect-all botocore s3_dlp_redteam_gui.py
```

8.2 Thread Safety

All AWS operations (uploads, cleanup, connection tests) run on background threads to keep the GUI responsive. UI updates (log, progress bar, results table) are safely scheduled on the main thread using tkinter's `after(0, callback)` mechanism — the correct approach for cross-thread GUI updates in Python.

8.3 No Data Leaves Without Authorization

The tool only communicates with AWS endpoints using credentials explicitly provided by the tester. No data is sent anywhere else. All test data is synthetic and generated locally.

9. Test Sign-Off

Role	Name	Date	Signature
Red Team Lead			

DLP Engineer			
Security Manager			
Test Environment Owner			

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